
Berkeley, CA | oliverhchang@berkeley.edu | (408) 203-9356

EDUCATION

University of California, Berkeley

B.S. Mechanical Engineering

Expected Graduation: May 2028 | GPA: 3.96

SKILLS

Technical Skills: Onshape, Solidworks, Fusion 360, ANSYS, MATLAB, Python, 3D Printing, CNC, Laser Cutting, Lathe, Mill

RELEVANT EXPERIENCE

Theoretical & Applied Fluid Dynamics Laboratory (TAFLAB)

May 2025 - Present

Undergraduate Researcher

- Engineered a complete wave energy harvesting system, leading the mechanical design, fabricating a multi-stage transmission, and integrating sensors through Arduino to consistently capture performance data.
- Prototyped 3 transmission architectures (geared, pulley, sprocket), designing waterproof enclosures, and testing jigs.
- Developed a physics-based model in Python to analyze system dynamics, using sensor data to quantify damping coefficients and energy loss, optimizing the design to successfully generate a consistent 0.5 W.

Formula Electric at Berkeley

Sept 2024 - Present

Chassis Mechanical Engineer

- Developed the team's first chassis FEA model in ANSYS, optimizing the spaceframe geometry to increase torsional stiffness
- Modeled the relationship between chassis torsional stiffness and vehicle handling, showed >1200 Nm/deg stiffness required for < 3% LLTD error and ensure predictable suspension tuning.
- Validated the FEA model by designing & building a torsional stiffness test rig, correlating physical data to simulation results with a <15% margin of error.

UC Berkeley's Space Technology and Rocketry (STAR)

Sept 2024 - Present

Airframe & Propulsion Engineer

- L2 Rocket: Designed rocket layout in OpenRocket, fabricated the airframe and recovery systems, and integrated avionics. Assembled, soldered, and programmed custom flight electronics for altitude, pressure, and temperature data. Successfully launched to 2,944 ft apogee.
- Liquid Engine: Applied DFM to manufacture 9 custom injector and combustion chamber components; used GD&T to ensure strict precision and tolerances for CNC/mill/lathe for successful static hot fire.

Fremont High Robotics FRC Team 3501: Firebots

2020 - 2024

VP of Mechanical Design

- Led a 6-person team to design a competition robot with Onshape that placed in the top 15% of 3,000+ teams worldwide, introduced new rapid prototyping workflow, refining design with major overhauls, and oversaw team's technical drawings

PROJECTS

Bike Trailer: Used Onshape to CAD a collapsible bike trailer, reducing to ¼ of its original footprint for storage.

Combat Robot: Designed, wired, and fabricated multiple fully 3D-printed competitive 1lb robots for local competitions. Engineering a Offset Slotted-Link shuffler driverbase mechanism utilizing a crank and slider design.

WORK EXPERIENCE

Scouts BSA Summer Camp Staff

June - July 2022, 2023, 2024

Kitchen Staff, Aquatics Staff, Handicraft Director

- Developed curriculum and instructed over 400+ scouts in first aid, lifesaving, woodworking, and technical skills.

HONORS & AWARDS

Autodesk Design & Make Ambassador

ASME 2025 Cadathon, 1st Place